

Math Sec 1

1. Let $S = \{z : 3 \leq |2z - 3(1 + i)| \leq 7\}$ be a set of complex numbers.

Then $\min_{z \in S} \left| \left(z + \frac{1}{2}(5 + 3i) \right) \right|$ is equal to:

- (A) $\frac{1}{2}$ (B) $\frac{5}{2}$
(C) 2 (D) $\frac{3}{2}$

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2. Let the mean and variance of 8 numbers $-10, -7, -1, x, y, 9, 2, 16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively.

Then the mean of 4 numbers $x, y, x + y + 1, |x - y|$ is :

- (A) 10 (B) 9 (C) 11 (D) 12

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3. Number of solutions of $\sqrt{3} \cos 2\theta + 8 \cos \theta + 3\sqrt{3} = 0$, $\theta \in [-3\pi, 2\pi]$ is:

- (A) 0 (B) 3 (C) 4 (D) 5

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4. If α and β ($\alpha < \beta$) are the roots of the equation $(-2 + \sqrt{3})(|\sqrt{x} - 3|) + (x - 6\sqrt{x}) + (9 - 2\sqrt{3}) = 0$, $x \geq 0$, then $\sqrt{\frac{\beta}{\alpha}} + \sqrt{\alpha\beta}$ is equal to:

- (A) 10 (B) 11 (C) 8 (D) 9

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5. The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \dots + \frac{{}^{100}C_{100}}{101}$ is :

- (A) $\frac{2^{100}}{100}$ (B) $\frac{2^{100}}{101}$
(C) $\frac{2^{101}}{100}$ (D) $\frac{2^{101}}{101}$

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6. A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

- (A) 36 days (B) 42 days (C) 24 days (D) 30 days

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7. Let $f(x) = \int \frac{(2-x^2) \cdot e^x}{(\sqrt{1+x})(1-x)^{3/2}} dx$. If $f(0) = 0$, then $f(\frac{1}{2})$ is equal to:

- (A) $\sqrt{3}e - 1$ (B) $\sqrt{2}e - 1$
(C) $\sqrt{2}e + 1$ (D) $\sqrt{3}e + 1$

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8. The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are $(1, 6, 3)$ and $(4, 9, \alpha)$ respectively and C is at a distance of 10 units from B. The area (in sq. units) of $\triangle ABC$ is :

- (A) $15\sqrt{13}$ (B) $20\sqrt{13}$
(C) $5\sqrt{13}$ (D) $10\sqrt{13}$

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9. Let $f(x) = \begin{cases} \frac{ax^2+2ax+3}{4x^2+4x-3}, & x \neq -\frac{3}{2}, \frac{1}{2} \\ b, & x = -\frac{3}{2}, \frac{1}{2} \end{cases}$ be continuous at $x = -\frac{3}{2}$. If $f \circ f(x) = \frac{7}{5}$, then x is equal to:

- (A) 1 (B) 2 (C) 1.4 (D) 0

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10. Let the line $y - x = 1$ intersect the ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$ at the points A and B. Then the angle made by the line segment AB at the center of the ellipse is:

- (A) $\frac{\pi}{2} - \tan^{-1}(\frac{1}{4})$
(B) $\frac{\pi}{2} + \tan^{-1}(\frac{1}{4})$
(C) $\pi - \tan^{-1}(\frac{1}{4})$
(D) $\frac{\pi}{2} + 2 \tan^{-1}(\frac{1}{4})$

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11. The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is :

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{18}$
(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

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12. A rectangle is formed by the lines $x = 0, y = 0, x = 6$ and $y = 4$. Let the line L be perpendicular to $3x + y + 6 = 0$ and divide the area of the rectangle into two equal parts. Then the distance of the point $(\frac{1}{2}, -5)$ from the line L is equal to :

- (A) $3\sqrt{10}$ (B) $\sqrt{10}$
(C) $2\sqrt{5}$ (D) $2\sqrt{10}$

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13. The sum of all possible values of $n \in \mathbb{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $(1 + x^2)^2 (1 + x)^n$, are in arithmetic progression is :

- (A) 7 (B) 12 (C) 9 (D) 3

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14. Let $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$.

Then $(\vec{a} - \vec{b}) \cdot \vec{d}$ is equal to:

- (A) 4 (B) 2 (C) -2 (D) -4

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15. Let the direction cosines of two lines satisfy the equations : $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$.

Then the cosine of the acute angle between these lines is :

- (A) $\frac{10}{3\sqrt{38}}$ (B) $\frac{10}{7\sqrt{38}}$
(C) $\frac{10}{\sqrt{38}}$ (D) $\frac{20}{3\sqrt{38}}$

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16. Let α and β respectively be the maximum and the minimum values of the function

$$f(\theta) = 4 \left(\sin^4 \left(\frac{7\pi}{2} - \theta \right) + \sin^4(11\pi + \theta) \right) - 2 \left(\sin^6 \left(\frac{3\pi}{2} - \theta \right) + \sin^6(9\pi + \theta) \right),$$

Then $\alpha + 2\beta$ is equal to:

- (A) 3 (B) 4 (C) 5 (D) 6

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17. Among the statements :

I: If $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$,
then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{3}{2}$, and

II : If $\begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x-3 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, the
 $p^2 = 196q^2$,

- (A) both are false (B) only I is true
(C) both are true (D) only II is true

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18. Let $y = y(x)$ be the solution of the differential equation $x^4 dy + (4x^3 y + 2 \sin x) dx = 0$, $x > 0$, $y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to:

- (A) 64 (B) 92 (C) 72 (D) 81

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19. Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$. Let R be a relation on A defined by xRy if and only if $2x + y \leq 2$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l + m + n$ is equal to :

- (A) 35 (B) 32 (C) 34 (D) 33

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20. Let the domain of the function $f(x) = \log_3 \log_5 \log_7 (9x - x^2 - 13)$ be the interval (m, n) . Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ have eccentricity $\frac{n}{3}$ and the length of the latus rectum $\frac{8m}{3}$. Then $b^2 - a^2$ is equal to :

- (A) 9 (B) 7 (C) 5 (D) 11

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Math Sec 2

21. Let $|A| = 6$, where A is a 3×3 matrix. If $|\text{adj}(3 \text{adj}(A^2 \cdot \text{adj}(2A)))| = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, then $m + n$ is equal to ____.

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22. Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0$, $x = \frac{3\pi}{2}$, and the x -axis be A . Then, $A + A^2$ is equal to ____.

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23. From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $a - b \geq 10$ is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to ____.

24. The number of 4-letter words, with or without meaning which can be formed using the letters PQRPRSTUVI is ____.

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25. Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x ((f(t))^2 + (f'(t))^2) dt$. Then the mean value of $f(\log_e(1)), f(\log_e(2)), \dots, f(\log_e(625))$ is equal to ____.

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Phy Sec 1